PROJECT FALCON-01 — ILLUSTRATED ELECTRONICS WIRING DIAGRAM

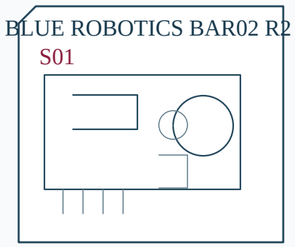
ACTUAL-PART DRAWING STYLE · ADVISER-ALIGNED PHASE 1 BASELINE · POWER / SENSOR / TELEMETRY INTERFACES

REFERENCE — VERIFY BEFORE WIRING

A — FIELD SENSORS

BLUE ROBOTICS BAR02 R2

S01

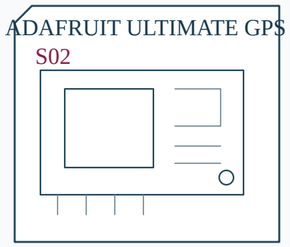


J2 · JST-GH-4
1 RED +3V3_SENSOR
2 GREEN I2C_SCL
3 WHITE I2C_SDA
4 BLACK GND

Pressure sensing face exposed to water; rear electronics remain dry.

ADAFRUIT ULTIMATE GPS

S02

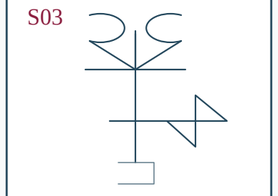


J3 · MODULE HEADER
PAD 2 +3V3_SENSOR
PAD 3 GND
PAD 4 GPS_RX ← GPIO17
PAD 5 GPS_TX → GPIO16

TX/RX cross at the controller.

SPARKFUN WEATHER METER

S03



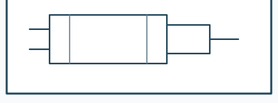
J7 · ANEMOMETER
1 +3V3 · 2 PULSE · 3 GND

J8 · WIND VANE
1 +3V3 · 2 VANE · 3 GND

Vane → ADS1115 A0 / 3.3 V divider

BLUE ROBOTICS CELSIUS R2

S04



J9 · DEPLOYMENT CANDIDATE
+3V3 · I2C_SCL · I2C_SDA · GND

Verify exact harness/address before PCB freeze.

DS18B20: bench-only alternative.

B — ESP32 CONTROL & SIGNAL WIRING

I2C SDA/SCL

GPS UART · GPIO16/17

WIND_PULSE · GPIO25

WIND_VANE → ADS1115 A0

CELSIUS I2C · VERIFY

ESP32-DEVKITC V4

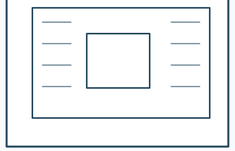
U3



USB / 5 V commissioning only
3.3 V logic · common documented GND

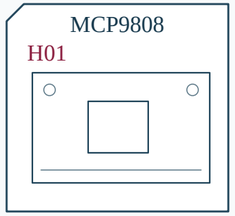
ADS1115 ADC

U4



MCP9808

H01



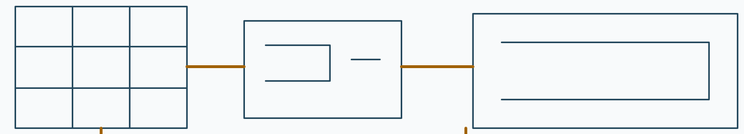
I2C 0x48 / 0x18
MCP9808: mount away from heat

CONTACT SWITCH · GPIO33



BUZZER + MOSFET · GPIO27
Exact buzzer/driver remains TBD

C — POWER & HEALTH MONITORING



P1 · 2 × 30 W SOLAR

P2 · MPPT

P3 · 12.8 V LiFePO4

INA260 SOLAR · 0x41
H03

INA260 BATTERY · 0x40
H02

HIGH-CURRENT PATHS
stay off carrier headers
Logic header only:
3V3 · GND · SDA · SCL

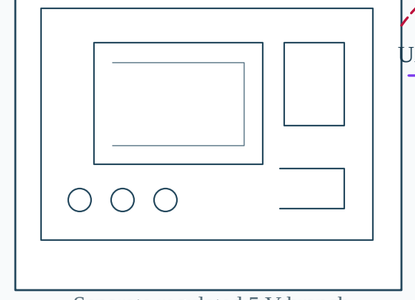
FUSED INPUT → REVERSE POLARITY / TVS → 5 V BUCK → 3V3 LDO

TP: VBAT · +5V · +3V3 · GND · validate ratings by load test

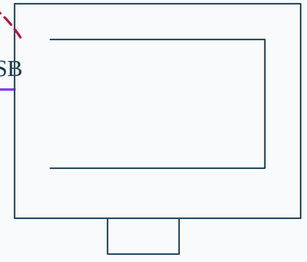
D — TELEMETRY / SHORE LINK

WAVESHARE SIM7600G-H 4G HAT

C02



UART/USB



SHORE BAY STATION
Database · pressure processing
AI prediction · dashboard · alerts

Separate regulated 5 V branch
Peak current and interface: TBD

E — CONTROLLED CONNECTOR SUMMARY / DRAWING NOTES

J2 BAR02	3V3 · SCL · SDA · GND	J6 MCP9808	3V3 · SCL · SDA · GND	J9 CELSIUS	Final I2C harness/address: VERIFY
J3 GPS	3V3 · GND · GPS_RX · GPS_TX	J7 ANEMOMETER	3V3 · WIND_PULSE · GND	LTE LINK	UART/USB and voltage levels: NOT FROZEN
J4/J5 INA260	3V3 · GND · SCL · SDA (logic only)	J8 WIND VANE	3V3 · WIND_VANE · GND	SERVICE	USB/UART for bench commissioning only

POWER · I2C · UART/USB · GPIO/ANALOG · CELLULAR

NOTES: 1) Pin numbers are provisional until exact purchased modules and connector orientations are verified. 2) Disconnect USB, battery and solar before rewiring. 3) No BNO085 or Orange Pi is part of the Phase 1 buoy harness. 4) KiCad schematic and verified harness register control fabrication—not this illustrated reference.

DRAWING	FALCON-BP-007	REV	P0
STATUS	REFERENCE / NOT FOR FABRICATION	DATE	2026-09-01